

Laiba Fatima

22K-5195

BSE-BB

Date: _____

Subject: _____

PPIT A3

1.

a) Year	Cash Outflow	Net cash flow
0	25000	-25000
1	800+200+400 = 1400	-1400
2	800+200+400 = 1400	-1400
3	800+200+600 = 1600	-1600
4	800+200+600 = 1600	-1600
5	800+200+600 = 3200 -5000	+3200

b) $NPV = \frac{\sum C_t}{(1+r)^t}$; where C_t = cashflow
 $r = 10\% = 0.10$

Year	Cash Flow	Discount factor	Present value
0	-25000	1.000	-25000
1	-1400	0.909	-1272.73
2	-1400	0.826	-1156.40
3	-1600	0.751	-1201.60
4	-1600	0.683	-1092.80
5	+3200	0.621	+1987.20

$NPV = -25000 - 27,736.33$

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c) The present value of costs exceeds the sale benefits as NPV is negative, so the company should not invest in delivery truck

2. Year	Cash Outflow	Net cash flow
0	15000	-15000
1	$300 + 250 = 550$	-550
2	$300 + 250 = 550$	-550
3	$300 + 250 = 550$	-550
4	$500 + 250 = 750$	-750
5	$(500 + 250) - 2500 = -1750$	+1750

Year	Cash flow	Discount factor	Present value
0	-15000	1	-15000
1	-550	0.926	-509.3
2	-550	0.857	-471.35
3	-550	0.794	-436.7
4	-750	0.735	-551.25
5	+1750	0.681	+1191.75

$$NPV = -15776.85$$

It is negative so don't buy the equipment.

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3 a) Year	Cash Flow	Discount Factor	Present Value
0	-18000	1	-18000
1	4000	0.893	3572
2	5000	0.797	3985
3	5500	0.712	3916
4	6000	0.636	3816
5	6500	0.567	3685.5

NPV = +974.50 (Positive)

b) Year	Cash Inflow	Cumulative
1	4000	4000
2	5000	9000
3	5000	14500
4	6000	20700
5	6500	27000

$$\text{Payback period} = 3 + \left(\frac{18000 - 14500}{6000} \right)$$

$$= 3.59 \approx 3.6 \text{ years}$$

c) It is viable, has positive NPV & under 4 years (less than 5)